



Team Project Software Engineering

Background

Your software development start-up has been awarded a contract to design and implement a prototype **operational information system for First Quantum Minerals**. The objective of this project is to simulate a real-world enterprise software development process, from requirements analysis through system implementation and documentation.

Task

Working in groups, you are required to **design, develop, and document** an operational software system that supports selected business processes of a large mining organization.

The system should be developed using an appropriate programming platform such as **Python, Java, or C#**, and must demonstrate sound application of software engineering principles.

Your project must include:

- Requirement's analysis and system specification
- System design using UML
- Implementation of core system features
- Error handling and exception management
- System testing and validation
- User and technical documentation

Deliverables

1. Operational Software System

- A functional prototype implementing key system features
- Developed using Python, Java, or C# (or another approved platform)

2. Technical and User Documentation Manual

- Minimum length: **50 pages**
- Format: Formal software manual
- Must include:
 - System overview and objectives
 - Requirement's specification
 - Architecture and design decisions
 - UML diagrams and explanations
 - Error handling and troubleshooting guide
 - Installation and user instructions
 - Testing and evaluation
- Referencing style: **APA**

3. UML Design Artefacts

A minimum of **four (4)** UML diagrams, for example:

- Use Case Diagram
- Class Diagram
- Sequence Diagram
- Deployment or Component Diagram

Instructions:

- Assignments submitted after the first 24 hours deadline will have 5% deduction.
- Assignments submitted after the 2nd 24 hours will be an additional 5%.
- Third 24hours a further 5% deduction.
- Should your assignment submission go beyond Day 3, 0% will be awarded for the assignment, but the assignment must still be submitted in order to pass the course.